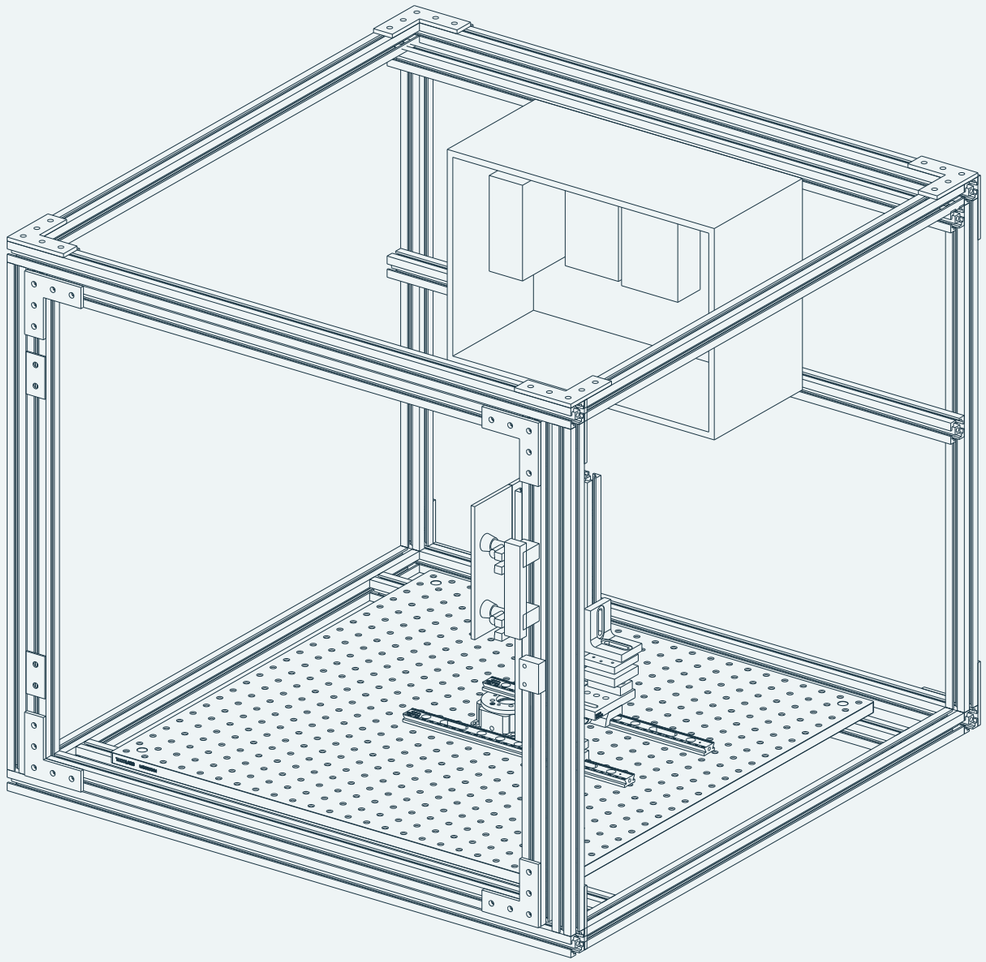
01 / Receive and segregate the purchased kit

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
Assembly parts register; vendor drawings; measurement traveler

1. Confirm exact revision and all part suffixes before unpacking. Segregate imperial #6-32 hardware from M4. Verify the R451P-XY is the factory plain configuration, with no springs or micrometers.

2. Check supplier-cut lengths and squareness. Record extrusion, panel and component measurements against the drawings. Do not correct a wrong part by drilling, filing, tapping or machining it.

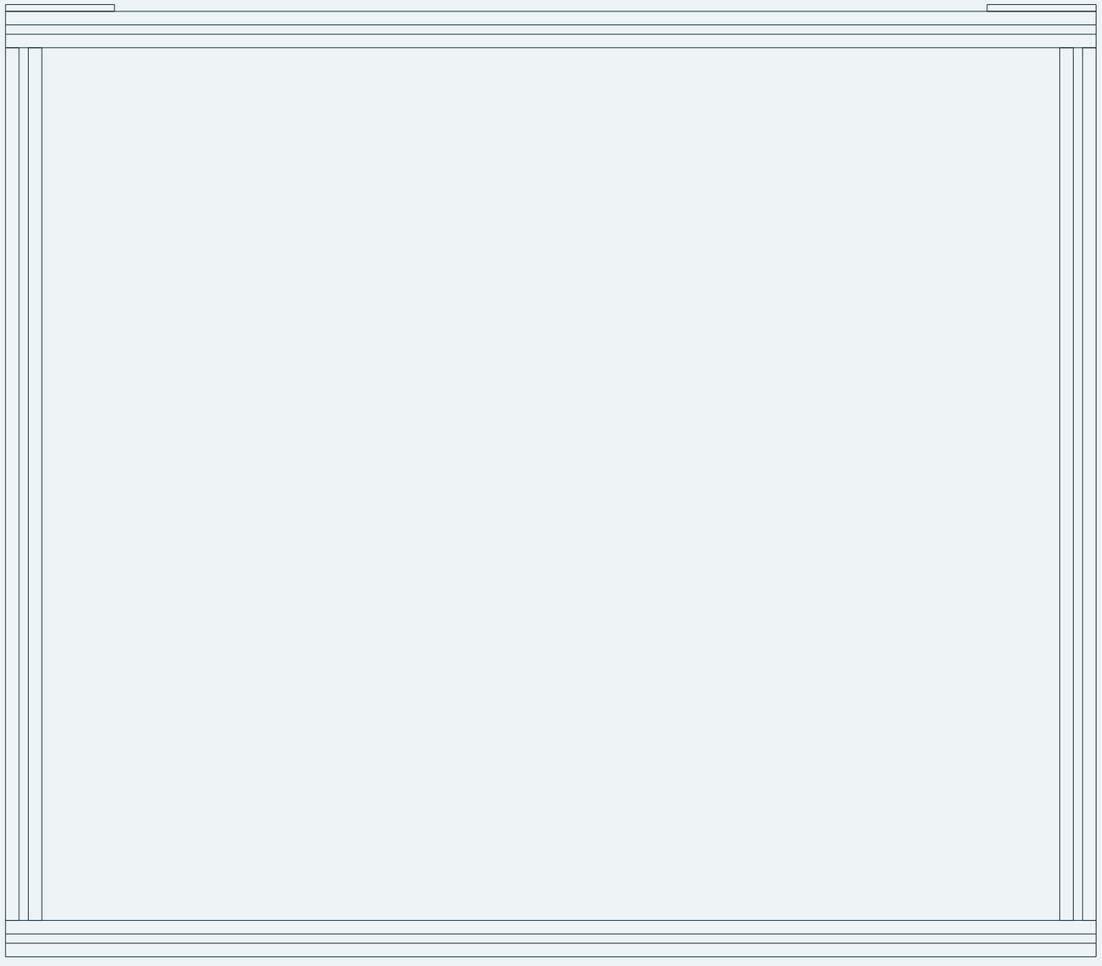
3. Identify original vendor CAD versus reconstructed envelopes in the model. Keep the original data sheets with the job traveler.

ACCEPTANCE
Record serial numbers, received quantities and drawing revisions. Photograph identification labels. No purchased substitute is accepted on appearance alone.

HOLD / VERIFY
This is a development assembly sequence. Close the design holds listed in the release register before issuing a shop build release.

02 / Prepare the commercial support bench

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
C150 workbench; C151 anchoring candidates; frame base

1. Use the purchased welded workbench on a level floor. The intended 1524 x 914.4 mm steel top provides a continuous floor below the guard; a frame standing over an open edge does not.

2. Center the 762 mm frame footprint in depth. The nominal front and rear edge allowance is 76.2 mm. Keep the operator side clear and locate facility air and factory power supplies outside the motion area.

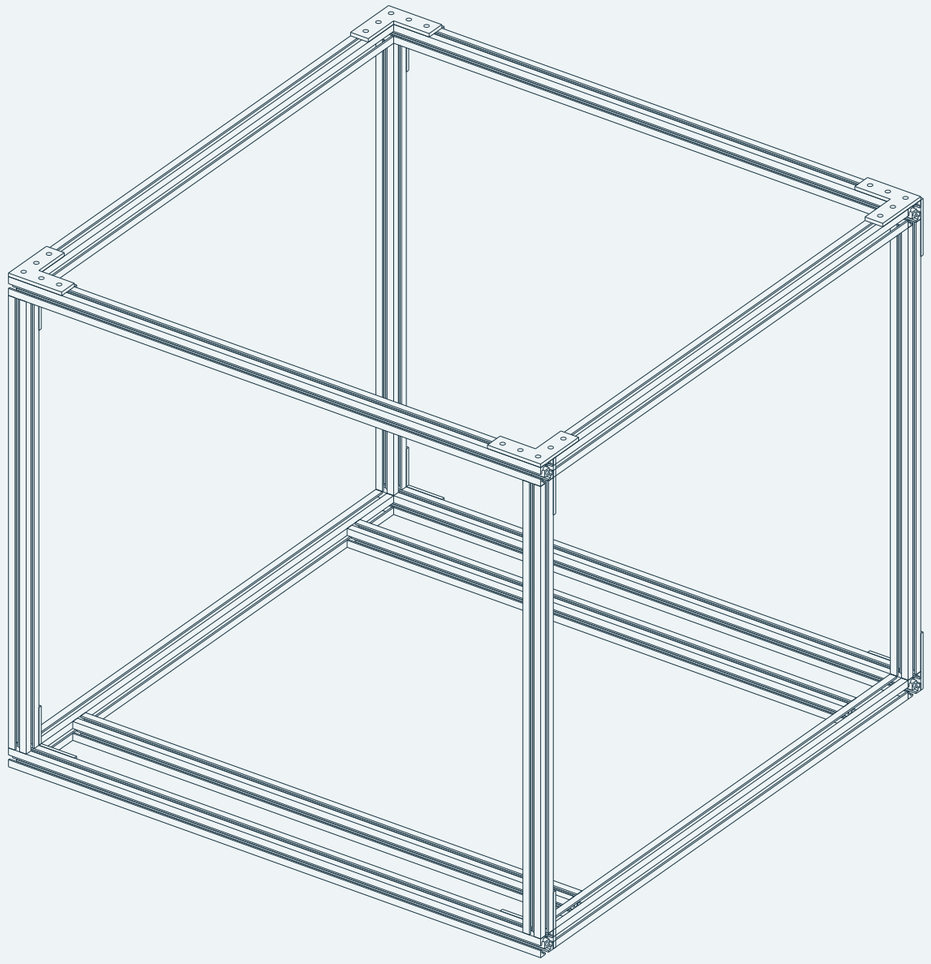
3. The proposed four stock clamps must reach 88.9 mm from the worktop edge to a frame-slot center. Confirm the actual workbench underside and clamp jaw geometry before purchase.

ACCEPTANCE
The base must resist measured operating and recovery loads without sliding, lifting or crushing an extrusion wall. Inspect restraint after empty cycling.

HOLD / VERIFY
M05: the stock clamp and restraint interface is not yet released. Do not rely on machine weight or the workbench load rating as an anchoring calculation.

03 / Assemble and square the lower frame

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
C101 x2; C102 x2 perimeter plus x2 supports; C112/C113 base joints

1. Lay two 762 mm members and two 711.2 mm members on the flat bench. Position the shorter members between the long members to obtain a 762 x 762 mm outside footprint.

2. Fit the purchased 4119 corner brackets using their matched screw/T-nut hardware. Keep fasteners loose enough to square the frame; all joining uses existing slots.

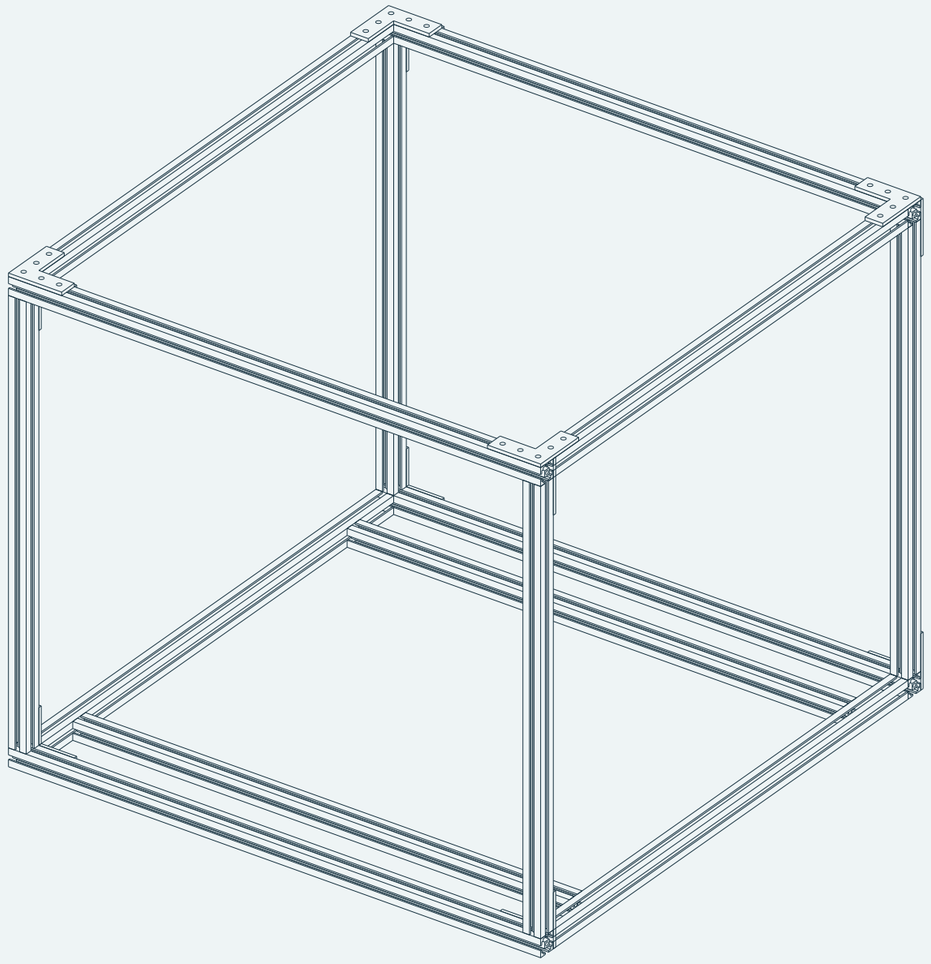
3. Place the two 711.2 mm breadboard support members with their centerlines at Y=-275 and +275 mm. Their top faces are Z=25.4 mm. Preserve access for the board mounting nuts.

ACCEPTANCE
Check opposite outside dimensions and equal diagonals. A design target is diagonal difference at most 0.5 mm before final tightening; record actual values. Confirm every bracket clamps rather than bottoms its screws.

HOLD / VERIFY
M04/M02: final bracket placements and all fastener stacks must be approved. Use the supplied bracket torque specification.

04 / Capture fixed panels and close the frame

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
C103 x4 posts; C101 x2 and C102 x2 top rails; C110 frame plates; C120/C121 panels

1. Set the four 609.6 mm posts on the base. Install the side and rear panels into their facing slots before adding the last members that close each opening.

2. Fit the upper 762/711.2 mm rectangle. The overall frame height is 660.4 mm. Capture the roof panel during closure; do not drill retaining holes through it.

3. Install the 12 purchased 4151 frame plates, five 3321 fastener sets each. Put the four front upright plates on the inside/back face at Y=-355.6 mm so they do not obstruct the door sweep.

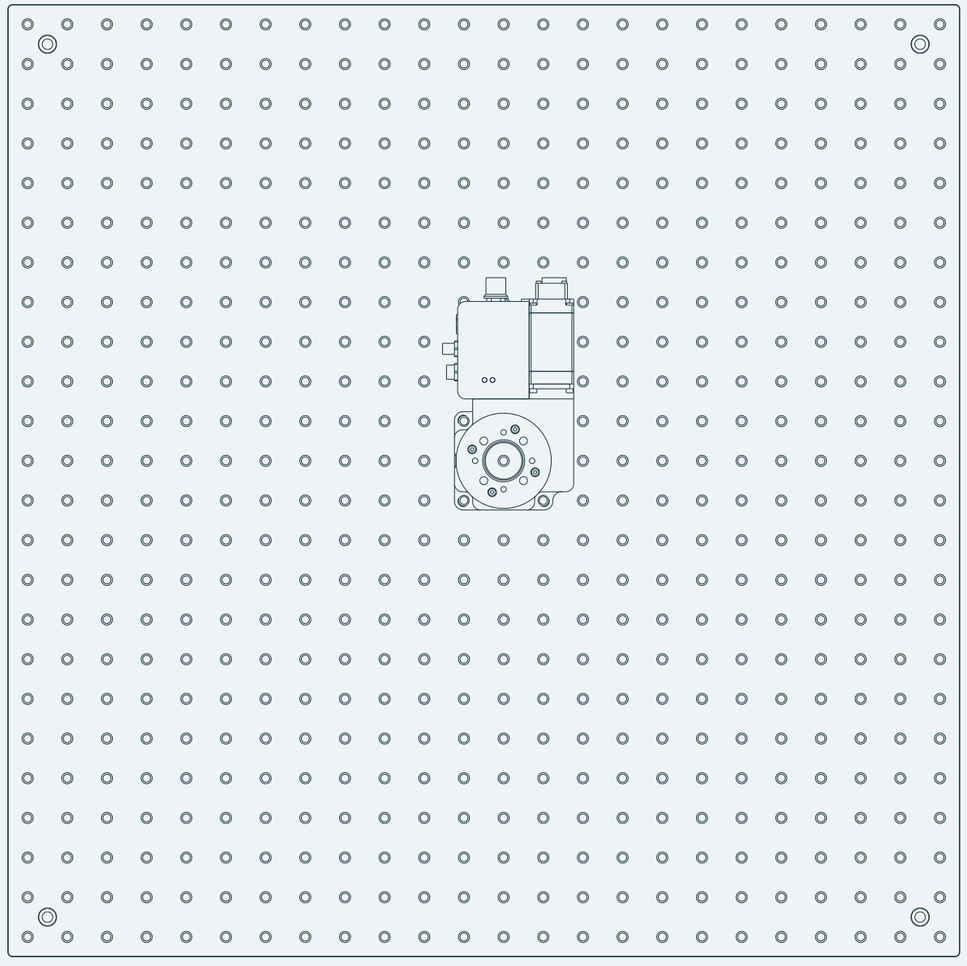
4. Insert the selected panel gasket only after the panel sits freely in its slots. Nominal edge capture is 6 mm, not the full slot depth.

ACCEPTANCE
Check plumb, squareness and panel retention on all four edges. Leave no unsupported opening at the base. Verify plate fasteners do not intersect panel edges.

HOLD / VERIFY
M03: 2116 gasket compatibility and retention with the actual panel thickness must be confirmed before these cuts are released.

05 / Install the breadboard and rotary stage

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
C130 MB6060/M; C131 X-RSW60A-E03; four 3873 nuts; fastener schedule

1. Center the 600 x 600 mm breadboard on the lower supports. Its bottom is Z=25.4 mm and its top is Z=38.1 mm. Align its corner mounting holes with the support T-slots.

2. Check each counterbored mounting screw's seat-to-slot projection and nut engagement. Tightening against a slot floor is not acceptable.

3. Mount the rotary stage at global axis X=12.5, Y=12.5 mm using the existing breadboard grid and manufacturer mounting pattern. Keep the stage base flat and the data/power connections accessible.

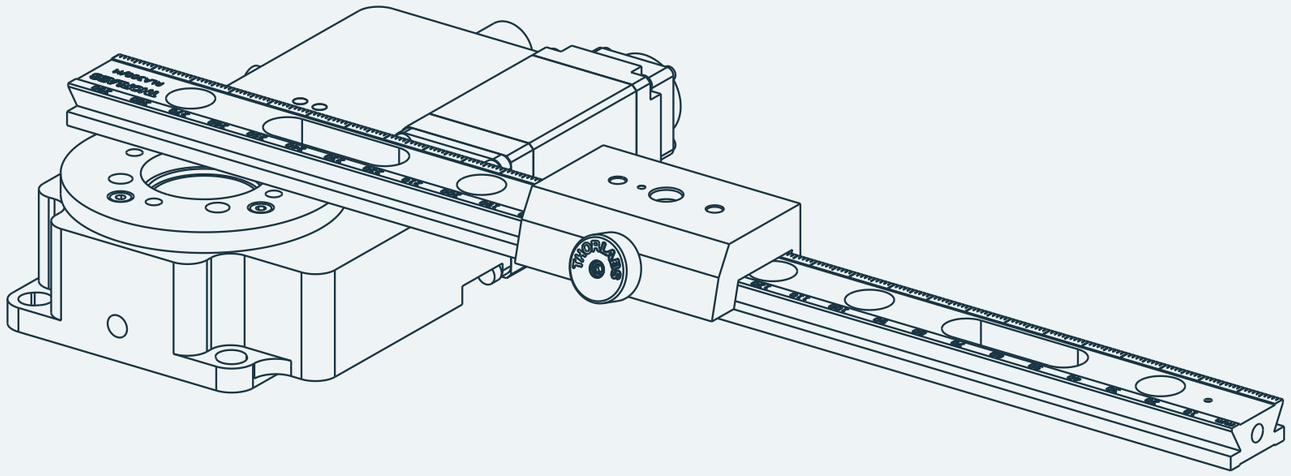
4. Do not energize or home the stage with fixture parts or a phone installed.

ACCEPTANCE
Confirm base seating, all manufacturer-required mounting screws, tool access and underside clearance. Record the actual axis location and cable exit direction.

HOLD / VERIFY
M02: board screw engagement and final stage-base screw count must be closed against the vendor drawing.

06 / Install the rotating adjustment rail

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
C132 RLA300/M; C133 RC2/M; two M6 x 8 stage-top screws

1. Orient the 300 mm rail along local X. Its center is X=112.5, Y=12.5 mm relative to the stage axis. Its rear end is X=-37.5 mm; do not center the rail on the stage.

2. Use the existing end-region rail holes over the selected 25 mm rotary-table row. Two M6 x 8 screws project nominally 5.46 mm below the rail seat. M6 x 10 is rejected for this interface.

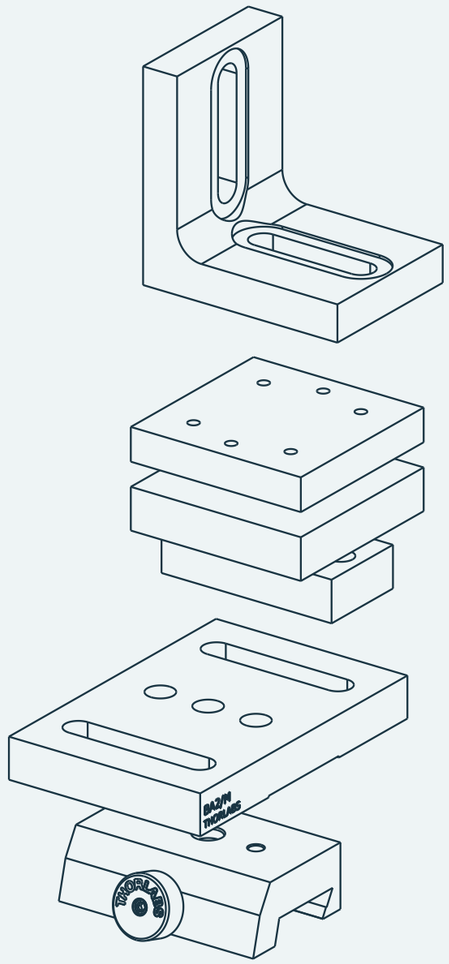
3. Slide on the RC2/M carrier. Set its nominal local X center to 110.099 mm and lock it. This carrier provides setup adjustment only; it is not the floating compliance bearing.

ACCEPTANCE
Verify two seated screws, usable blind depth, unlocked slide travel during setup and positive carrier locking for operation.

HOLD / VERIFY
Keep the stage unpowered. The rear-rail clearance is essential: the earlier centered arrangement collided with the fixed grip.

07 / Build the purchased floating XY stack

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
C134 BA2/M; C137 SD1; C138 R451P-XY; C136 AB90; C140/C142/C143 hardware

1. Fit BA2/M to the locked moving carrier with two M4 x 14 screws and SD1 adapters. The nominal stack is 8.509 mm, leaving 5.491 mm engagement in the carrier.

2. Place the R451P-XY base on the BA2 outer slot. Its two existing #6 clearance holes are 33.3248 mm apart. Through-bolt with the selected #6 hardware; no slot enlargement is permitted.

3. Attach the AB90 horizontal leg to the XY carriage with the two #6-32 x 5/8 screws and proposed counterbore adapters. Keep the stage near mid-travel while tightening.

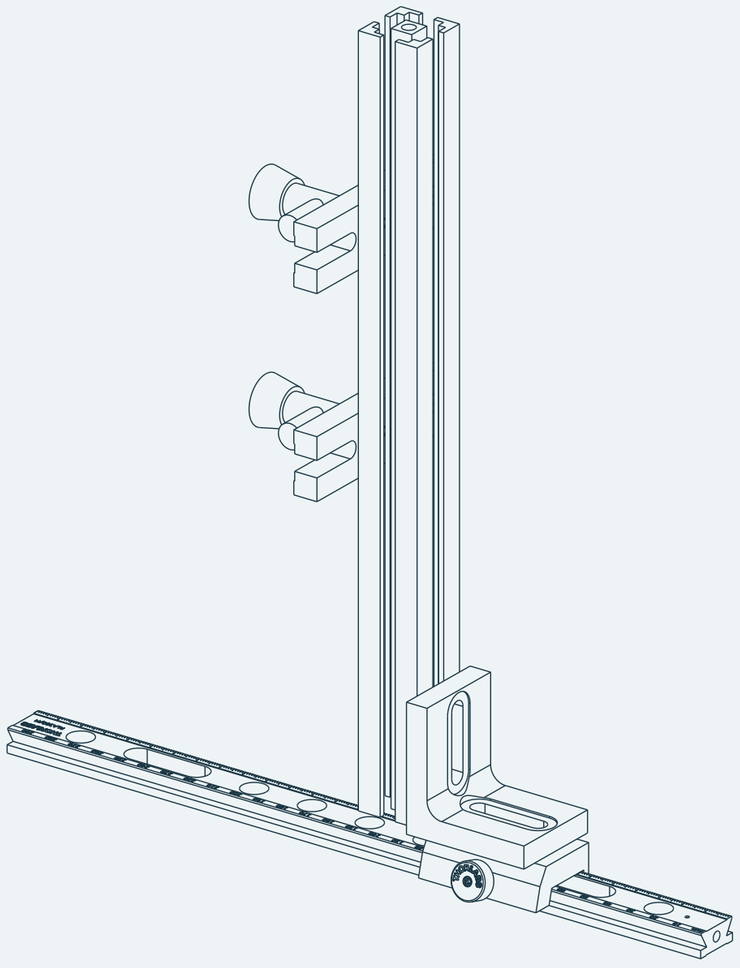
4. The XY unit remains free in both axes during cycling. The RC2 stays locked. Do not add a return spring or lock the passive stage to make setup easier.

ACCEPTANCE
Verify full catalog travel, no fastener interference, nut bearing area, positive seating and low breakaway drag. Reserve at least 2.5 mm from each end of the 25.4 mm total stroke.

HOLD / VERIFY
M02: #6 head support on SD1 and the through-bolt nut/washer stack remain engineering holds. The exploded spacing is illustrative, not an installation dimension.

08 / Assemble the fixed grip support

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
Second RLA300/M and RC2/M; AB90; 300 mm 1010 mast; C141 and T-slot hardware

1. Mount the stationary 300 mm rail to the breadboard using its existing holes. Nominal local rail center is X=50, Y=-50 mm relative to the rotary axis.

2. Set and lock its RC2/M carrier. Fasten AB90 with two M4 x 16 screws and SD1 adapters; nominal projection is 5.713 mm into the 6 mm usable carrier depth.

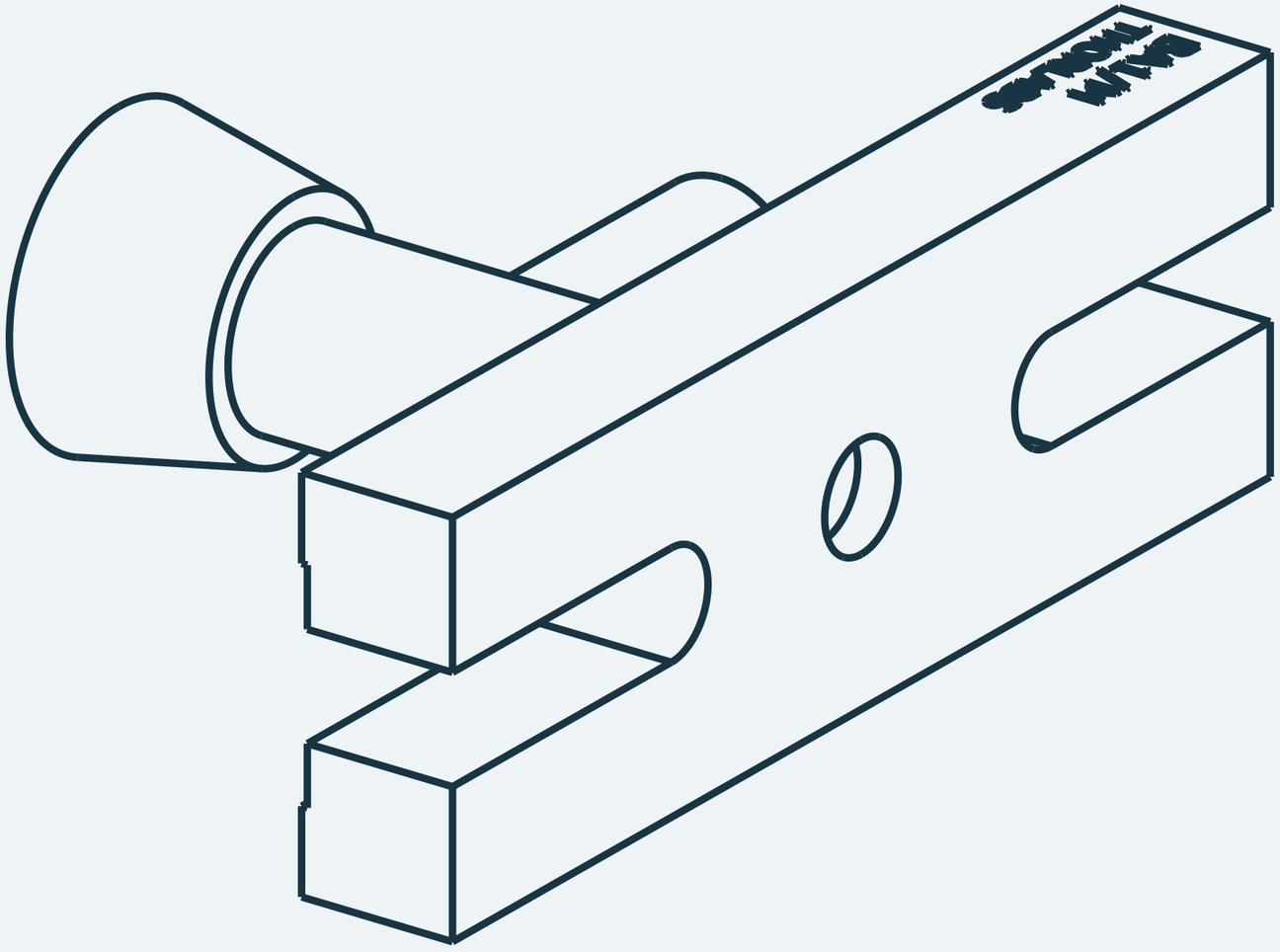
3. Attach the 300 mm mast to the AB90 vertical slot through two 3873 M6 T-nuts. Keep the cup-mounting face toward the phone and retain adjustment until phone setup.

HOLD / VERIFY
The fixed side is rigid after setup. It must not follow the rotating rail or be used as a flexible restraint.

ACCEPTANCE
Check mast plumb and rotation resistance. Confirm screw tips clear the slot floor and bracket faces remain seated.

09 / Install the four outside-surface cup assemblies

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
C135 BA1/M x4; C302 ESH holders x4; C301 ESS cups x4; C144 screws

1. Insert the M6 holder screw from the rear of each BA1/M central counterbore before attaching the plate to a mast. Its nominal projection is 5.048 mm; measure actual usable blind thread first.

2. Attach each plate through its two outer slots to mast T-nuts. Nominal cup contact elevations are local Z=160 and 240 mm; these are surrogate setup positions, not measured phone dimensions.

3. Install the purchased cups on the holders according to Festo instructions. Orient tube exits away from the closing display gap.

4. Repeat on the 200 mm moving mast. No plate, screw, cable or tube may sit between the two display faces at natural closure.

ACCEPTANCE
Inspect cup lips, threads, coating contact and plate seating. Record received holder depth and the actual cup-center positions after phone measurement.

HOLD / VERIFY
Q01: actual phone geometry and surface compatibility must be measured before contact positions are accepted.

10 / Assemble and hang the front guard door

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
C104/C105 door extrusion; C122 panel; four C110 plates; C116-C119 hardware; C202/C203

1. Assemble the door from two 606.425 mm stiles and two 643.7252 mm crossbars. Capture the 655.7252 x 567.625 mm panel before tightening the last member.

2. Hang the door on two 2086 hinges centered at Z=150 and 510 mm. Closed outside edges are X=-354.0252 to +340.5 and Z=26.9875 to 633.4125 mm.

3. Mount the handle at right-stile center X=327.8, Z=440 mm. Its holes are 75.006 mm apart.

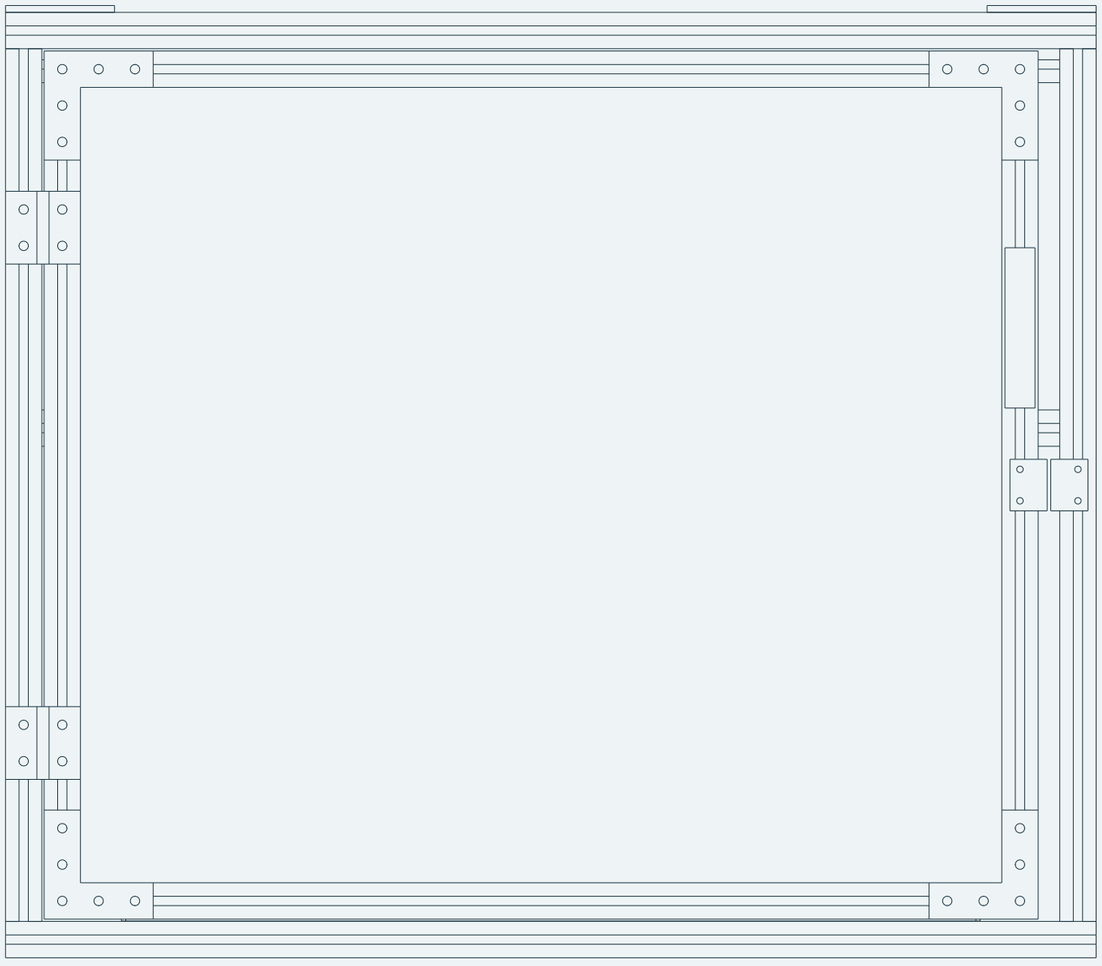
4. Mount BNS and BPS with their mounting-line centers at X=368.3 and 327.8, both Z=330.2. Each hole pair is 22 mm apart; face the marked active surfaces toward each other.

ACCEPTANCE
Nominal sensing gap is 2.5 mm. Verify alignment through repeated closure and opening. Use a separate catalog stop; neither switch housing may absorb door impact.

HOLD / VERIFY
M03: latch/stop, gasket retention and the 15.1 mm latch-side opening require completion and access/stopping-time validation.

11 / Mount the controls enclosure and operator stations

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
C106 rails; C220-C223 Altech parts; C201/C206/C207 modules; C204/C205 stations; C210/C211 Pi

1. Install the horizontal cabinet support rails just in front of the rear posts, centerline Y=342.9 mm. Shift their elevations to the supplied mounting tabs; do not drill the enclosure or extrusion.

2. Use the factory-knockout enclosure, purchased mounting tabs, hinge kit and precut DIN rails. Place the SC10 vertically with its ventilation clearances maintained.

3. Keep the E-stop and physical reset outside the guard, reachable without entering the motion area. Use purchased brackets and existing station mounting holes.

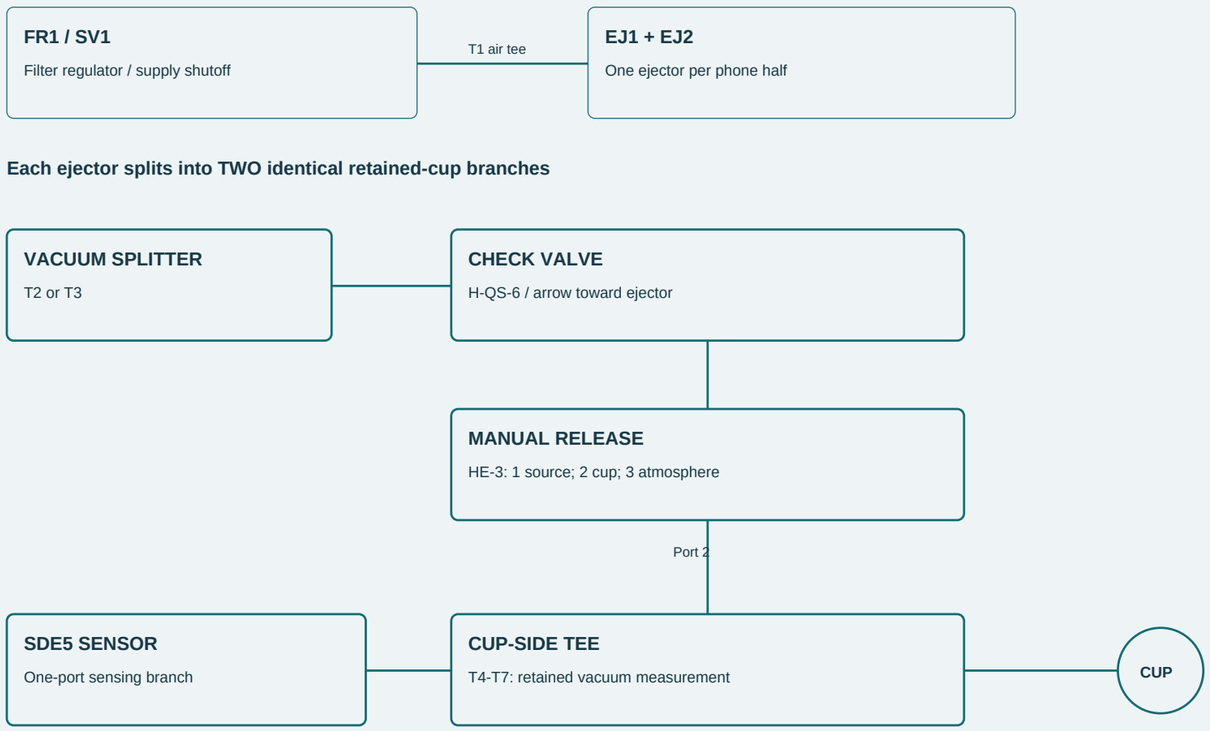
4. Mount the Pi in its purchased DIN case and preserve USB, Ethernet and microSD access. Keep factory power supplies outside the cabinet.

ACCEPTANCE
Check lid travel, terminal access, gland sealing, bend radii and all hardware against the actual factory boss drawing.

HOLD / VERIFY
M04/E02: detailed mount stacks, terminal layout and gland order codes are incomplete. The CAD cabinet contents are dimensioned envelopes, not a released cabinet build.

12 / Install the pneumatic hardware

Purchased components only | Assembly sequence and acceptance



Repeat x4. Release only after both phone halves are supported.

ACCEPTANCE

Verify the circuit with a hand vacuum pump, supported sample and individual leak tests before connecting compressed air for motion.

PARTS / TOOLS

C303-C313; regulator bracket and nut; purchased mount hardware

1. Install the regulator upright with accessible gauge and drain. Fit its G1/4-to-6 mm adapters; do not interchange NPT and G threads.

2. Install one ejector per phone half, four cup check valves, four individual release valves and four sensing tees. Connect the one-port SDE5 switches only to the retained cup side.

3. Orient each check valve for airflow from cup toward ejector. Label each release valve HOLD/RELEASE and each branch V1-V4.

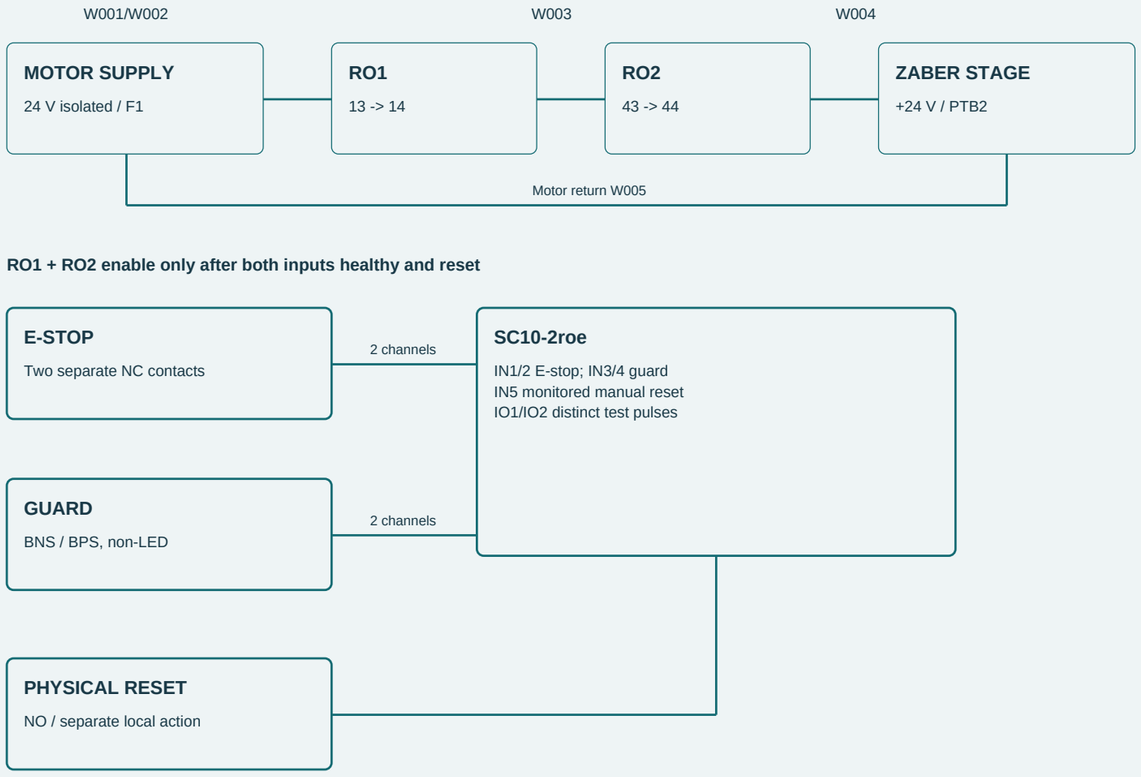
4. Use the labeled hose allowances in pneumatic-installation.md. Route and measure before final cutting; maintain at least 50 mm bend radius in moving loops as a design allowance.

HOLD / VERIFY

P01/P02: regulator, ejector, valve and sensor mount stacks plus dynamic routing remain open. A tidy schematic is not evidence that the hose sweep clears.

13 / Wire the independent stop circuit

Purchased components only | Assembly sequence and acceptance



PARTS / TOOLS
C201-C205; motor/control supplies; fuses; terminal system

1. Disconnect all supplies. Label every conductor at both ends using the electrical schedule. Keep motor and control supplies separate.

2. Route motor +24 V through F1, RO1:13-14 and RO2:43-44 in series. Route the auxiliary permit through RO1:23-24 and RO2:53-54 to physical DI1.

3. Wire the two E-stop and two guard channels separately. For the non-LED BNS, black/blue and white/brown are contact pairs; brown is not a general +24 V supply lead.

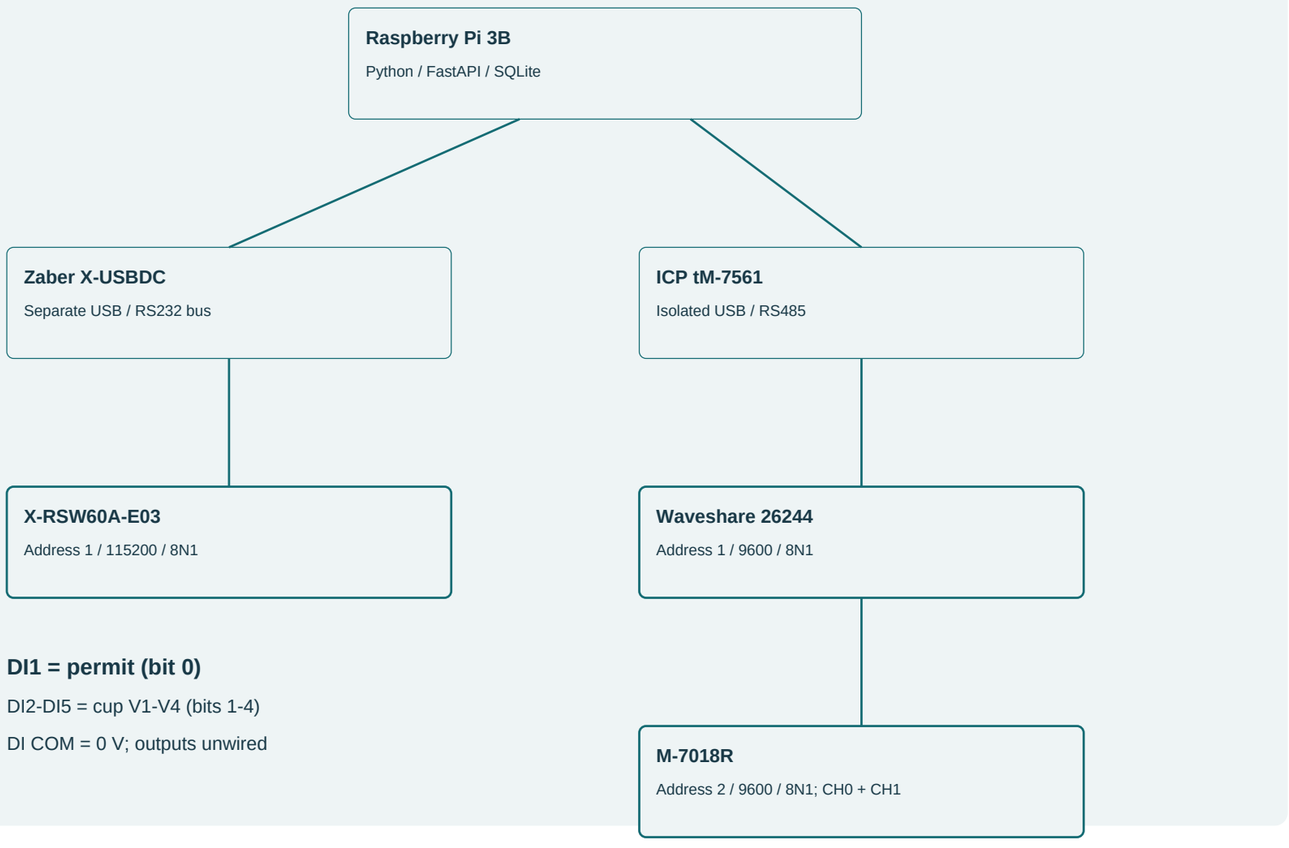
4. Wire the physical NO reset to the approved SC10 reset input. Export and review the actual Banner configuration and matching wiring report before energizing.

ACCEPTANCE
Continuity-test each channel independently, inspect polarity and conductor protection, and prove no alternate stage power path exists.

HOLD / VERIFY
E01-E04: connector compatibility, final terminal kit, converter map and Banner configuration remain design release gates.

14 / Connect process sensors and serial buses

Purchased components only | Assembly sequence and acceptance



PARTS / TOOLS
C206-C209; four SDE5 cables; two SA1-K probes; X-USBDC

1. Connect physical DI2, DI3, DI4 and DI5 to the four individual vacuum switch outputs. DI1 is permit. DI COM goes to 0 V for the selected PNP inputs.

2. Connect Type K probes to CH0 and CH1 with correct polarity. Keep the phone probe clear of cups and the hinge; keep ambient sensing away from exhaust and warm electronics.

3. Connect the Zaber factory RS232 USB lead on its own port. Daisy-chain the RS485 converter, Waveshare and M-7018R with addresses 1 and 2 on that bus.

4. Configure baud rate, channel type, engineering units and enabled channels with vendor utilities. Record stable USB adapter names and all hardware serial identities.

ACCEPTANCE
Read actual raw values; disconnect every probe and vacuum switch individually. Confirm the expected fault and latency. No output terminal is connected to a cup-release valve.

HOLD / VERIFY
E03/Q03: termination/reference wiring and real hardware behavior require documented verification.

15 / Install and exercise the Raspberry Pi service

Purchased components only | Assembly sequence and acceptance



Physical E-stop and guard operate independently of this stack.

PARTS / TOOLS
Pi 3B; purchased supply and storage; software source and Pipenv lock

1. Install the locked runtime using Pipenv on a supported 64-bit Linux system with Python 3.12 or later. CAD and document tooling run on the workstation, not the Pi.

2. Start python -m foldbench.revc in default simulation mode. Use a generated API key, one worker and persistent storage. Check /health, then authenticated /v1/status.

3. Exercise a simulated cycle, pause, resume and stop. Inject each supplied simulation fault. Verify incomplete cycles are not counted and a process restart never resumes a run.

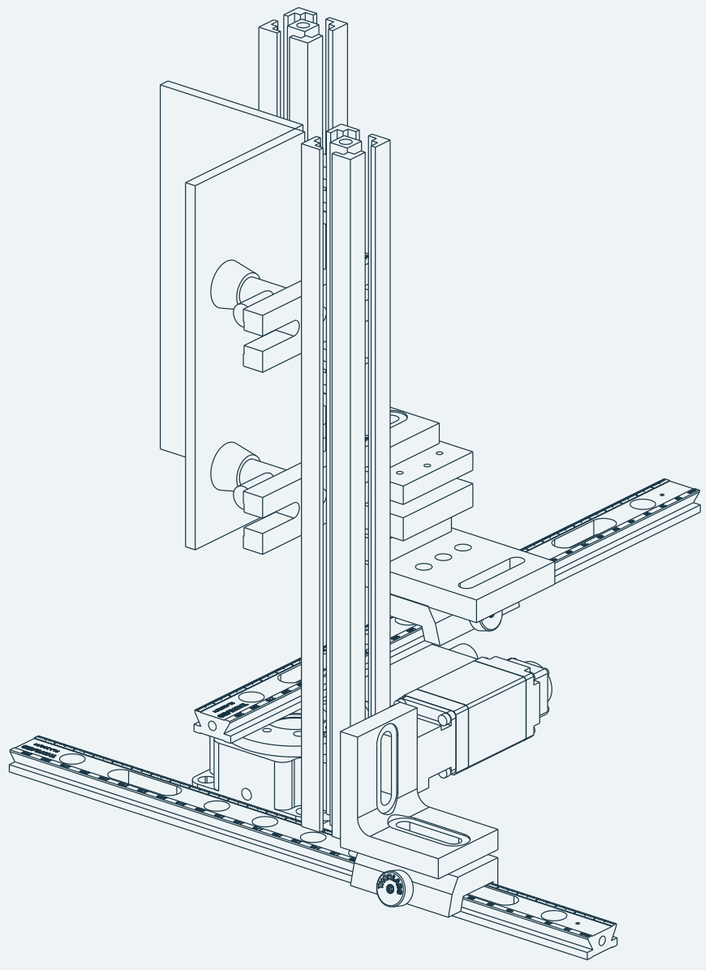
4. Install the supplied systemd service only after local simulation succeeds. Keep the equipment API on a trusted network; the public revision gallery is a separate static site.

ACCEPTANCE
Use the supplied OpenAPI contracts and tests. Store credentials and the commissioning file outside Git. Confirm simulation-only routes are absent in hardware mode.

HOLD / VERIFY
No code test proves physical stop performance or cup retention. Hardware mode stays locked until the commissioning evidence is completed.

16 / Qualify motion with the fixture empty

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
Completed assembly; dial indicator; gauges; stop-time measurement; commissioning record

1. Complete all design holds first. With no phone fitted, establish the stage reference using the vendor tool and verify motion direction.

2. Move slowly through the entire intended stroke. Measure passive XY position, minimum clearances, hose drag and any joint movement. Repeat in both directions.

3. Test guard opening, E-stop, each process input fault, communication loss and host restart. Record actual stopping time, last movement and recovery behavior.

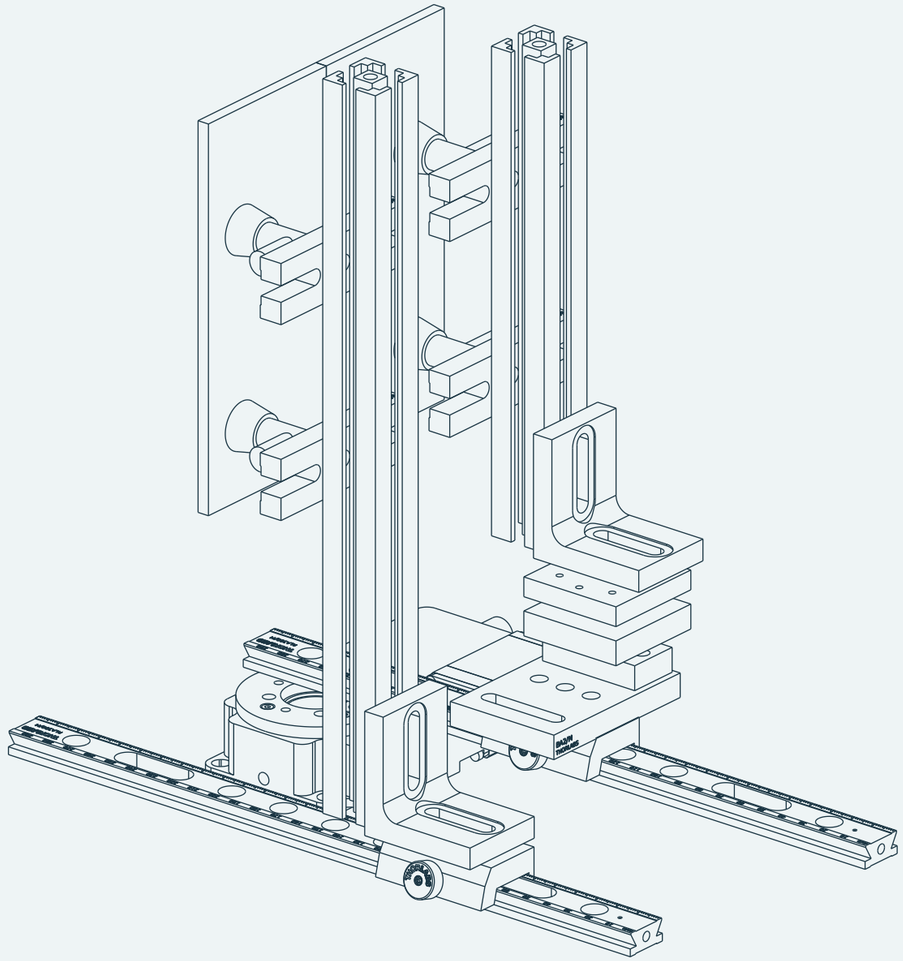
4. Confirm the stage has no alternate power source and reset never begins motion. A worm-driven stage may retain load after power removal; define the recovery procedure.

ACCEPTANCE
Record results and photographs for every failure injection. A sampled CAD sweep and a passing API response are not substitutes for these tests.

HOLD / VERIFY
Q02/Q03: no powered phone testing until empty-fixture acceptance is signed.

17 / Measure and qualify the actual phone

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
Phone measurement traveler; representative sacrificial specimen; four cup assemblies

1. Measure the actual phone's open/closed geometry, hinge path, mass, camera bumps and suitable outside contact areas. The 70 x 160 x 5 mm CAD leaves are only illustrative.

2. Set both locked carriers and all cup positions to the measured specimen. Support both leaves while applying vacuum. Verify each cup reaches its individually qualified threshold.

3. Determine allowable hinge/contact loads and verify combined XY stage loads and life. The bench has no direct force or torque sensor; motor current is not a phone torque measurement.

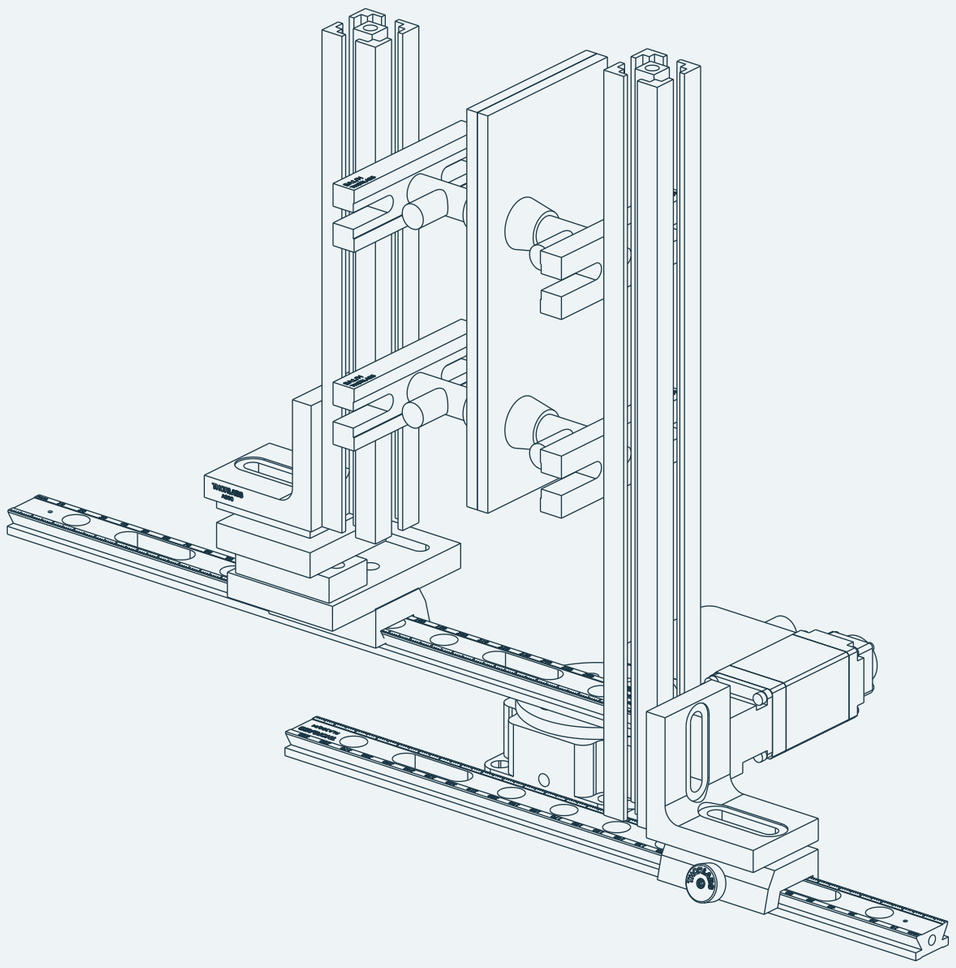
4. Test supply-loss retention, surface marking, peel resistance and supported manual release on the sacrificial specimen before installing the target phone.

ACCEPTANCE
Record cup locations, vacuum thresholds/hysteresis, retained-vacuum duration, XY margins and approved operating speed/current.

HOLD / VERIFY
Q01/P03: the shop measures the actual phone as authorized by the project owner. Do not infer product dimensions from a name or marketing video.

18 / Accept full natural closure and hand over

Purchased components only | Assembly sequence and acceptance



CAD review view - not to scale

PARTS / TOOLS
Qualified specimen; final commissioning file; as-built drawings and records

1. Establish the phone's natural closed endpoint without fixture force. The two display faces must reach that state with no hardware, hose or cable between them.

2. Commission the motor endpoints to the accepted physical range. Verify the compliant XY stage remains away from both ends of travel. Never overdrive a hard endpoint to force a nominal 180 degree value.

3. Run the agreed supervised acceptance cycles, inspect contact marking and joint security, and verify the cycle count against observed complete close/open pairs.

4. Deliver the as-built terminal report, hardware identities, approved limits, source revision, recovery procedure and inspection interval with the signed traveler.

ACCEPTANCE
A complete cycle ends open. Pause and resume require deliberate commands; an interrupted run stays ended after reset. Archive all acceptance evidence with the revision.

HOLD / VERIFY
The present C-0.3 package has open design holds. This handover step describes the required end state, not a claim that a physical bench has been accepted.